



Data colonialism and indigenous languages in AI: a critical review of existing initiatives and their struggles with data sovereignty

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Abstract

This article critically reviews recent initiatives to employ artificial intelligence (AI), particularly large language models (LLMs), for the revitalization of Indigenous languages. Structured by geographical contexts, the analysis includes Irish Gaelic (Europe), Māori (Aotearoa/New Zealand, Oceania), Guaraní (Paraguay/Bolivia, South America), and Inuktitut (Canada, North America). Applying a theoretical framework grounded in data colonialism and Indigenous data sovereignty, the article examines the key achievements in different regional endeavors, as well as investigate how government-led projects and Big Tech collaborations across these diverse contexts navigate (or fail to navigate) issues of data extraction, community consent, cultural representation, and ownership. Through this lens, the article identifies specific ethical pitfalls as well as commendable practices that either reproduce colonial dynamics or empower Indigenous communities. This critique emphasizes regional and contextual nuances, arguing that authentic community agency and rigorous adherence to Indigenous data sovereignty principles are vital to ensuring ethical AI practices and meaningful linguistic revitalization.

Keywords Indigenous language revitalization · Large language models (LLMs) · Data colonialism · Indigenous data sovereignty · Ethical AI · Indigenous community agency

1 Introduction

Artificial intelligence (AI) language technologies have surged in capability and prominence, raising both hopes and concerns for endangered Indigenous languages. On one hand, generative AI and large language models (LLMs) offer new tools for documenting and revitalizing threatened tongues. UNESCO estimates that around 40% of the world's 7000 languages are endangered, with nearly half possibly vanishing by 2100 (IBM Research 2021). In response, researchers and communities are exploring AI-driven solutions ranging from speech recognition to text generation to help preserve linguistic heritage. For example, with sufficient data, LLMs could be trained to produce content in endangered languages, assist education, and archive oral histories. Such developments promise a “digital renaissance” (Berez 2013; Galla 2016; Sk 2024): the revival,

rejuvenation, or renewed flourishing of culture, language, knowledge, or arts through the use of digital technology and media for minority languages, bridging them into modern platforms and encouraging new generations to learn and use them. On the other hand, the rise of AI brings significant risks of digital and data colonialism. The very forces that make AI powerful, such as massive data collection and globalized models, tend to favor dominant languages and cultures, often at the expense of smaller communities. Internet and AI technologies so far have reinforced the hegemony of English and other major languages such as French, Spanish, and German, marginalizing Indigenous languages in cyberspace. There is a real danger that AI could further entrench power asymmetry, as Big Tech companies harvest linguistic data from Indigenous communities without consent or benefit, and deploy AI tools that fail to respect local context or knowledge. Thus, alongside its promise, AI poses a threat of digital colonization (Coudry and Mejias 2019), where Indigenous languages become mere raw material for global AI systems, their speakers left with little control or reward.

With the unprecedented opportunities and profound risks for Indigenous language revitalization presented by AI, a tension vividly illustrated by recent global initiatives

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is observed. Ireland exemplifies this duality through two concurrent approaches: the research-driven development of UCCIX, the first open-source Irish Gaelic LLM fine-tuned on a small curated corpus to outperform larger multilingual models despite revealing vulnerabilities like “catastrophic forgetting” (Tran et al. 2024), and the state-backed *ArdIntleacht na Gaeilge* (literally “High Intelligence of the Irish Language”) project, which aims to deploy a comprehensive Irish-language AI assistant for public services by 2026, explicitly seeking to “safeguard Irish in the digital realm” (Caollaí, 2024). This juxtaposition of grassroots (i.e. bottom-up) innovation and institutional commitment (top-down) reflects a broader global pattern in AI deployment for linguistic revival. Across Indigenous contexts, divergent models emerge as well, including government-led initiatives prioritizing scalability yet risking centralized control, community-pioneered tools centering Indigenous governance while facing resource constraints (as demonstrated by Te Hiku Media’s 90% accurate Māori speech models) and Big Tech collaborations offering computational power but threatening data commodification. The Irish case thus anchors our critique of how AI navigates colonial legacies, framing this paper’s core question of whether such projects empower communities through data sovereignty or replicate extraction through technological solutionism. When in the Guaraní Mbya case in Brazil, project activities were suspended after the Tenondé-Porã community raised concerns about exposing their language and culture online and deliberated on whether and how the initiative should continue (Pinhanez et al. 2024), or Inuit technologists embedded communal control into Inuktitut tools, they affirmed non-negotiable principles of agency, data ownership, and cultural integrity. Rather than focusing on technical prowess alone, this paper argues that these principles determine AI’s impact by first dissecting Irish Gaelic projects (UCCIX and *ArdIntleacht na Gaeilge*) through the lens of data colonialism versus sovereignty, then juxtaposing them with Māori, Guaraní, and Inuktitut cases to map ethical pitfalls and commendable practices across the Indigenous AI landscape.

2 Theoretical framework

2.1 Data colonialism

The discussion is situated within two major theoretical frameworks of data colonialism and Indigenous data sovereignty (with a focus on the CARE Principles, in contrast to mainstream FAIR data principles). In the subsequent sections, these frameworks are applied to evaluate the Gaelic case studies and comparative examples. This paper then looks at the Irish Gaelic AI initiatives (UCCIX and *ArdIntleacht na Gaeilge*) by contextualizing these projects.

Subsequently, their opportunities and risks are analyzed through these theoretical lenses, helping us assess to what extent these attempts resist data colonialist patterns, uphold Indigenous data governance, and promote epistemic justice. The discussion is then broadened to Māori, Guaraní, and Inuktitut cases, examining how similar issues play out across different Indigenous contexts. Finally, the paper concludes with recommendations for ethically harnessing LLMs for endangered languages, aiming to maximize language revitalization benefits while minimizing harms of digital colonialism. Through this interdisciplinary analysis, the paper seeks to contribute to the discourse on how AI can be developed in a way that supports linguistic diversity and Indigenous rights, rather than undermining them.

The concept of data colonialism provides a critical foundation to understand how modern AI and data-driven technologies can perpetuate colonial patterns of extraction and domination. Coined by Couldry and Mejias (2019), data colonialism refers to “the massive appropriation of social life through data extraction”, a process that represents “a new colonial move” in which human experiences become a resource to be mined for economic value (Couldry and Mejias 2019). In traditional colonialism, foreign powers seized land, labor, and resources; in data colonialism, corporations and states seize personal and collective data on an unprecedented scale. Crucially, the underlying drive remains the same: to acquire “territory” and resources (now, information and human expressions) from which capitalist enterprises can extract profit. This framework highlights that even though the context and methods have changed (digital networks instead of naval fleets), the power asymmetry and dispossession inherent in the AI system for preserving Indigenous language echo those of historical colonialism. The result is a new social order where those who control data reap the benefits, while individuals and communities become sources of raw data subjected to external control.

Applying this idea to AI language technologies, one can see how Indigenous languages might be at risk of data colonialism. AI systems, especially LLMs, are typically trained on enormous corpora scraped from the web or other sources with little transparency. In many cases, Indigenous language data (texts, audio recordings, translations, etc.) are harvested and aggregated into AI training datasets without the knowledge or consent of the communities who created them (Tu 2025). For example, public texts in Irish or Māori might be scooped up by web crawlers to improve a tech giant’s translation engine, with the resulting AI product then sold back to consumers. This may and can include the Indigenous community as a service, often with no compensation or benefit flowing to that community. This practice has been likened to “digital extraction” and indeed “echoes historic patterns of dispossession and exploitation”, as one policy analysis noted in the context of Indigenous data being used to train

commercial algorithms without consent (Tu 2025). In a data-colonial scenario, Indigenous communities lose control of their linguistic resources, which are absorbed into proprietary AI models, much as colonized peoples lost control of land and knowledge to imperial powers.

2.2 Indigenous data sovereignty: ownership and existing practices for protection through CARE principles vs FAIR principles

With data colonialism in view, an immediate question arises: whose data, precisely, is being colonized? In positive law, no community “owns” a natural language as an abstract code. What law protects are the material instantiations and infrastructures that render language computable: (i) expressive works (texts, sound recordings) via copyright; (ii) databases/corpora via copyright in selection/arrangement and, in the EU, a *sui generis* database right over substantial extraction or re-use; and (iii) contract/licensing terms that govern access and downstream uses (e.g., machine-learning (ML) training and hosting). Consequently, the analytically sound frame for Indigenous claims is not property in the language itself but stewardship and governance over datasets, annotations, alignments, and trained models that encode collective knowledge and risk. Even when materials are publicly available or state-produced, a data-colonial lens shows how indiscriminate harvesting and proprietary enclosure of these assets reproduce colonial asymmetries, converting lived practices and community archives into extractable inputs for external value chains. Where public-access obligations apply (e.g., open government records), a layered governance approach can reconcile openness with community custodianship of sensitive corpora and model pipelines, foregrounding consent, reciprocity, provenance, and benefit-sharing rather than asserting ownership over the abstract code of the language.

In light of the above and to counter the threat of data colonialism, Indigenous scholars and communities advance the framework of Indigenous Data Sovereignty (IDS), a framework under which the regional attempts at a digital renaissance of Indigenous language are measured against. The IDS was proposed with the idea that Indigenous Peoples have the right to control the collection, ownership, and application of data about their members, cultures, and lands. In essence, IDS extends the political notion of sovereignty into the realm of information: it asserts that data relating to Indigenous communities *should* be governed by those communities, according to their values and for their collective benefit (Carroll et al. 2020). This principle arises from the recognition that data, including language as data, are not just abstract bits of information, but are intimately tied to community identity, well-being, and self-determination. When outsiders unilaterally control Indigenous data, they repeat

the injustices of colonialism; conversely, when Indigenous groups direct how their data are used, they can empower sustainable development, cultural preservation, and innovation on their own terms.

A concrete expression of Indigenous Data Sovereignty is found in the CARE Principles for Indigenous Data Governance, formulated in 2019 by the Global Indigenous Data Alliance and related networks (Carroll et al. 2020). CARE is an acronym for **C**ollective Benefit, **A**uthority to Control, **R**esponsibility, and **E**thics (Carroll et al. 2020). These principles were developed in consultation with Indigenous communities worldwide to complement the mainstream data management principles known as FAIR: **F**indable, **A**ccessible, **I**nteroperable, **R**eusable (Carroll et al. 2020). The key distinction is that while FAIR principles focus on making data maximally open and shareable—goals that align with scientific efficiency, CARE principles focus on people and purpose, emphasizing that data use should align with the rights and needs of Indigenous Peoples. In other words, CARE asks who benefits from data, who has authority over it, who is responsible for its impacts, and whether its use is ethical in the specific cultural context.

When it comes to AI and LLMs, applying the CARE principles could transform how Indigenous language projects are conducted. For instance, instead of scraping specific Indigenous text from the internet without permission, a CARE-oriented project would collaborate with the respective communities to curate datasets, with clear agreements that the community retains authority over how the model or dataset is used. This approach is exemplified by initiatives like Te Hiku in New Zealand, which negotiated data licenses ensuring that Māori voice recordings would remain under Māori community control and be used only in ways that benefit te reo Māori. Similarly, the IBM Research project with the Nheengatu and Guaraní communities in Brazil explicitly had Indigenous collaborators lead and set the terms of the project, pledging not to reuse the data elsewhere without community consent (IBM Research 2021). Such practices operationalize Indigenous data sovereignty by building consent and community governance into AI development. They also highlight tensions with the conventional open-data mindset. For example, IBM researchers found that open-sourcing the tools for Indigenous language AI is easier than open-sourcing the data, since the latter raises questions of who owns a language and how to protect community rights. The CARE framework would argue that, in cases of Indigenous languages, data should not be treated as freely open if doing so contravenes the source community’s interests – a nuance that the FAIR principles alone would overlook.

In policy terms, IDS and the CARE principles are gaining traction as essential to ethical AI involving Indigenous peoples. They align with earlier protocols like Canada’s OCAP principles (Ownership, Control, Access, Possession) from

First Nations data governance (Tu 2025), reinforcing the notion that communities should dictate the terms of data use. However, the integration of these principles into AI development is still limited. National AI strategies rarely mandate Indigenous data governance, and many academic or commercial AI projects proceed without engaging the relevant communities. A shift toward Indigenous data sovereignty in AI would mean rethinking design and evaluation metrics: success would not only be about model accuracy, but also about community acceptance, mutual benefit, and cultural integrity.

3 Case studies and discussion

The following case studies typify the central problem explored in this paper: how to leverage LLMs and AI for Indigenous language revival without falling into the trap of data colonialism or cultural harm. Fine-tuning LLMs for a language like Irish Gaelic sits at a crossroads between opportunity and risk. On one side lies the opportunity to reinvigorate the language through expanding its digital presence, creating tools for learners and native speakers, and normalizing its use in technology. On the other side lies the harm if done inappropriately, which readily includes loss of community control over linguistic data, AI systems that misrepresent or commodify the language, or technologies that inadvertently privilege outside interests over the Indigenous community's needs. The Irish example encapsulates the application of AI in safeguarding Indigenous languages, while demonstrating a broader tension applicable to many Indigenous and minoritized languages today.

It is note-worthy that across cases, *who* leads an Indigenous-language AI project strongly shapes both ethics and outcomes. Government-led initiatives typically combine preservation with public-service mandates and national cultural policy (as the Ireland case will show). Platform/Big-Tech deployments tend to optimize service at scale, with limited consultation before and after deployment, raising familiar data-colonial concerns about model control, training-data provenance, and benefit-sharing. By contrast, community-led models anchor aims in sovereignty and guardianship (as in the Māori case). The same technical systems (automatic speech recognition (ASR), machine translation (MT), LLMs) can be steered to different ends depending on who defines success.

3.1 Irish Gaelic (UCCIX and ArdIntleacht na Gaeilge): a national initiative

Irish Gaelic (Gaeilge) illustrates the dual predicament faced by Indigenous languages implementing AI for preservation. Irish is the national and first official language of Ireland, yet

it is classified by UNESCO as a “Definitely Endangered” language (UNESCO 2010), meaning it is no longer being learned naturally by most children. While census figures count roughly 1.8 million Irish citizens with some ability in Gaelic, the number of daily fluent speakers is much lower than that. In effect, Irish Gaelic is an Indigenous language of its own land now at risk, sustained by community efforts and policy support, but threatened by the dominance of English. This makes it a compelling case study of a minority language navigating the AI era.

Generations of Anglophone dominance have left Irish with a fraught sociolinguistic status: spoken as a first language by a minority, primarily in Gaeltacht regions, and facing ongoing intergenerational attrition. Nevertheless, Irish remains central to Ireland's cultural heritage and legal framework, enjoying protection under the 20-Year Strategy for the Irish Language 2010–2030 and official status in the EU. The Irish government has increasingly recognized that technology will be pivotal to the language's future. In 2022 it launched a comprehensive Digital Plan for the Irish Language, aiming to integrate Irish into “today's digital society” (Department of Tourism, Culture, Arts, Gaeltacht, Sport and Media, Government of Ireland 2023a) and enable the Irish-speaking community to “build and revitalise the language” through technology (Department of Tourism, Culture, Arts, Gaeltacht, Sport and Media, Government of Ireland 2023b). This plan backs research on speech synthesis, speech recognition, machine translation and other tools to ensure Irish “keeps pace with major European languages” in the digital age (Department of Tourism, Culture, Arts, Gaeltacht, Sport and Media, Government of Ireland 2023b). An initial €1 million investment in 2023 supported these efforts, with a commitment to open-source outcomes so that any public or private provider can deploy Irish language tech freely. Early foundational work on Irish language NLP laid important groundwork; for example, Uí Dhonnchadha and van Genabith (2006) developed a finite-state morphology and constraint grammar disambiguation, together with a part-of-speech tagger for Irish, one of several “considerable efforts...made over the past decade to develop natural language processing resources” for Irish. Such resources, including parsing tools and the “gaBERT” Irish neural language model, established a baseline digital infrastructure for Gaeilge.

Building on these foundations, new projects are leveraging LLMs to push Irish NLP into a modern era. A landmark effort is UCCIX, an open-source Irish LLM introduced in 2024 (Tran et al. 2024). UCCIX (shorthand for “Irish-eXcellence”) tackles the extremely low-resource nature of Irish by adopting a continued pre-training strategy on top of a pre-existing model. The team started with Meta's LLaMa-2 (13B parameters) and further trained it on Irish language data, but in a highly data-efficient way. Remarkably, they achieved

strong results “with only a fraction of the textual data typically needed” according to standard scaling laws (Tran et al. 2024). Since Irish has only hundreds of millions of tokens available for training, UCCIX introduced novel techniques to maximize learning from this small corpus through meticulous data curation and adaptive training regimes. The resulting model, at 13B parameters, outperforms even much larger general models on Irish-specific tasks by up to 12%. This is a significant achievement, demonstrating that a focused LLM can excel in a minority language context. In addition, UCCIX contributed new evaluation benchmarks¹ to enable rigorous testing of Irish language understanding and generation. These benchmarks fill a crucial gap for assessing model quality in Irish. Overall, UCCIX shows impressive proficiency in understanding and generating Irish text, validating that LLM technology can be adapted to serve a language of only approximately 170,000 daily speakers. Its creators explicitly frame it as a blueprint for other low-resource and Indigenous languages, with the dual goals of preserving Irish in the digital era and promoting Irish knowledge and culture globally. In this sense, UCCIX embodies the promise of LLMs for revitalization: it provides a state-of-the-art Irish language model that could power everything from machine translation and speech agents to educational tools—all with Irish at the core rather than an afterthought.

Alongside academic projects like UCCIX, the Irish government has initiated its own flagship AI effort, *Ard-Intleacht na Gaeilge* (literally “High-Intelligence of the Irish Language”). Announced in late 2024, *Ard-Intleacht na Gaeilge* is a public-sector project led by *Údaráis na Gaeltachta* (the Gaeltacht development authority) to create an Irish-centric AI system by 2026 (Caollai 2024). The vision for *Ard-Intleacht* is an AI assistant that can comprehend and respond in Irish across dialects, perform real-time Irish speech recognition and synthesis, handle translation between Irish and other languages, and crucially, imbue AI outputs with authentic Irish cultural context. In practical terms, this could mean Irish-speaking citizens interacting with government services through Irish voice commands, receiving AI-driven answers in their chosen dialect (Connacht, Munster, or Ulster Irish), or tourists getting culturally nuanced Irish-language content. The project was spurred by recognition that mainstream voice assistants and translation tools trained predominantly on English are flawed when it comes to Irish, often failing to capture Ireland’s cultural context or the nuances of the language. By investing in a bespoke Irish AI, the government aims to safeguard the position of Irish in the digital realm and ensure the bilingual public service goal by 2030 is bolstered by technology. The *Ard-Intleacht*

initiative has political weight behind it: it was highlighted in the first-ever National Plan for Irish Language Public Services 2024–2030, alongside the establishment of an advisory committee on the Irish language in the digital sphere. This committee includes key organizations and experts to guide digital Irish-language efforts, exemplifying a multi-stakeholder governance approach. The State’s role in *Ard-Intleacht* also brings funding and access to “the State’s existing pool of Irish language content, which includes archive audio and video” and text corpora (Caollai 2024). We see here a notable strategy: leverage public archives and media (much of it from RTE, TG4, etc.) as training data, and do so under public oversight.

The Irish Gaelic AI efforts are conscious of the issue of safeguarding data, but the application of an Anglophone LLM suggests that these maneuvers may be susceptible to influences of colonial or Anglophone ideologies. As a nationally recognized language, Irish occupies a somewhat different position than many Indigenous languages. The Irish state itself is a custodian of the language. Yet the concept of Indigenous data sovereignty is still highly relevant. Irish speakers (especially in Gaeltacht communities) are a minority with rights to control how their language and data are used. In the Irish context, this translates into questions about who owns the colossal amounts of text and speech being used to train Gaelic LLMs, and who decides on their applications. The UCCIX project, notably, received support from an Irish cloud computing company and national research grants, but the model and datasets were released openly. By open-sourcing the Irish LLM and benchmarks, the researchers ensured that the Irish public and scholars collectively benefit—a step showing awareness of open data principles and arguably with Indigenous data governance, since no single company can lock up the model or its underlying Irish corpus. The *Ard-Intleacht* project similarly indicates an intent to deploy under free licenses for public use, meaning the tools, such as speech recognition engines, would be freely available for anyone to integrate Irish language support into services. These choices help prevent a scenario where Irish data is extracted to train proprietary AI that Irish people then have to pay to use. It is a proactive guard against data colonialism, where human data is commodified. Nevertheless, Birhane (2021) argues that Western AI often imposes rigid, decontextualized systems that clash with the communal and contextual ways knowledge is held in Indigenous cultures. This tension is evident in Irish Gaelic too: an AI chatbot might be technically fluent, but if its values and conversational style are rooted in Anglophone internet culture which could subtly erode Gaelic worldviews or promote misrepresentations of Irish life. There is also the risk of dependency: if Irish initiatives rely heavily on foreign-developed infrastructure (e.g. using a closed-source API for speech synthesis), the community may lose self-sufficiency.

¹ Notably IrishQA, a question-answering dataset in Irish, and an Irish version of the MT-Bench test suite.

Finally, surveillance and commercialization concerns must be noted. AI tools like voice assistants could surveil Irish speakers if deployed without privacy safeguards, and culturally important Gaelic content (songs, stories, folklore) could be appropriated by AI companies to generate profit (e.g. AI-generated audio of sean-nós² songs sold as a service). These scenarios underscore that who builds and governs Gaelic AI is as important as what the AI does, and remain unaddressed issues in these two efforts.

3.2 Māori (te reo Māori, New Zealand), Guaraní (Paraguay/Bolivia), Inuktitut (Canada): practices in grounding AI in indigenous sovereignty

Similar to the Irish AI efforts, the Māori project also rejected Big-Tech, top-down hegemony. The Māori language has seen a renaissance in New Zealand, and Māori leaders have been at the forefront of developing AI tools grounded in cultural sovereignty. A standout example is the work of Te Hiku Media, a Māori-owned media organization that expanded from radio broadcasting into AI development. Te Hiku amassed a large archive of Māori audio over decades of producing radio and television content (Lee 2024). Rather than handing this valuable data to a tech giant, Te Hiku pursued a community-driven project called Papa Reo to create speech recognition and synthesis models by Māori, for Māori. According to Te Hiku's CTO, they adopted "ethical, transparent methods of speech data collection and analysis to maintain data sovereignty for the Māori people" in developing their ASR system (Lee 2024). The result was an ASR model that transcribes te reo Māori with 92% accuracy, rivaling commercial systems. Uniquely, it can handle code-switched speech (mixed Māori and English) with high accuracy as well, reflecting real usage in New Zealand. In practice, Te Hiku's project embodied Indigenous data governance, as the Māori community retained ownership of the speech corpus and the models, and they set the terms for any external collaboration. Indeed, Te Hiku partnered with NVIDIA to use its open-source NeMo toolkit and GPU hardware, but on Māori terms – the core mission remained "the preservation, promotion and revitalization of te reo Māori" rather than profit. The Te Hiku team even influenced industry norms, with Keoni Mahelona (Te Hiku's CTO) noting that they are "guiding the industry on ethical ways of using

data and technologies to ensure they're used for the empowerment of marginalized communities" (Lee 2024). This community-first ethic stands in contrast to extractive models. Instead of allowing Big Tech to scrape Māori words for a generic model, Te Hiku negotiated to lead the development and insisted on a *Kaitiakitanga* (guardianship) model for data. When Mozilla's Common Voice project sought Māori voice data, Te Hiku collaborated but required a data license that acknowledged Māori sovereignty over the recordings (Te Mana Raraunga n.d.). The success of the Māori ASR, now used in apps for language learning, transcription, and accessibility, demonstrates the power of Indigenous co-design. It also garnered global recognition³ (McCaull 2024). The Māori case illustrates that when a community controls its data and works with technologists as equal partners, AI can significantly amplify language revitalization efforts. At the same time, it flags ongoing challenges. Māori advocates have pointed out gaps in New Zealand law for protecting Indigenous data (e.g. no *sui generis* IP for data of national taonga/treasures). Thus, they are pushing not just for building models, but for legal and policy frameworks to protect what those models are built from. Overall, the Māori experience offers a template for other Indigenous communities: one that emphasizes sovereignty, cultural values, and long-term empowerment over quick commercial solutions, but with data ownership loopholes to fill.

In contrast to the earlier cases with smaller speaker bases, the case of Guaraní illustrates that even a widely spoken Indigenous language can be vulnerable to top-down interference by large technology companies. Guaraní is spoken by more than 6.5 million people (Pinhanez et al. 2024) in Paraguay (where it is co-official with Spanish) and in neighboring countries. One might assume that this large speaker base would translate into robust digital support. However, Guaraní has experienced sustained technological marginalization. Most Guaraní speakers are bilingual in Spanish, and historically language technology has prioritized the dominant Spanish. Local advocates note that despite an estimated 90% of Paraguay's population speaking and writing Guaraní, the language accounts for only about 20% of online content (Rising Voices 2020).

Compared with Spanish, natural language processing (NLP) and translation efforts for Guaraní have only recently gained traction. In 2022, Google added Guaraní to Google Translate as one of twenty-four under-resourced languages, using a zero-shot approach trained only on monolingual text given the scarcity of parallel corpora (Google Research 2022). Although this was the first appearance of Guaraní

² Literally translating as "old style" in Irish, refers primarily to a traditional form of unaccompanied singing characterized by ornate, richly expressive melodies, flexible rhythms, and highly personal emotional interpretations. Performances typically involve a single vocalist without any instrumental accompaniment, showcasing regional and individual stylistic variations. The tradition is deeply connected to Irish cultural heritage, particularly prominent in Gaelic-speaking areas such as Connemara, Donegal, and West Munster.

³ Te Hiku's CEO, Peter-Lucas Jones, made TIME's 2024 list of 100 most influential people in AI for championing Indigenous data sovereignty.

in a major translation application and a notable technical achievement, it was also largely an external, top-down inclusion. Google's public materials refer to "working with native speakers, teachers, and linguists," but present this as supporting a pre-existing corporate project, vaguely worded for keeping the development team "inspired with their passion and enthusiasm" rather than involvement in co-design or shared governance (MercoPress 2022). Despite the existence of Paraguayan institutions engaged in standardizing Guaraní,⁴ these announcements do not reference formal partnerships with government bodies, academic collaborators, or national institutions either. As Cooper, Heldreth, and Hutchinson argue, "NLP for Indigenous languages should primarily serve Indigenous communities," prioritizing "communities' values and experiences with respect to NLP projects", and the "expressed opinions of those communities around data governance" (Cooper et al. 2024). By contrast, Google's announcement foregrounds that the techniques introduced "supplement massively multilingual models with a self-supervised task to enable zero-resource translation," emphasizing methodological innovation and scale rather than community-defined aims (Google Research 2022). This stands in marked contrast to Ireland's government-led ArdIntleacht initiative and to Te Hiku Media's community-governed Māori model. From a data-colonialism perspective, one could further argue that publicly available Guaraní texts were appropriated to fuel proprietary models now accessed on corporate terms. The lack of co-governance and the unremunerated reuse of these texts echo Couldry and Mejias's account of data colonialism, in which extractive practices associated with "historical colonialism" are combined with the "quantification methods of modern computing" (Couldry & Mejias 2019; Gray 2023).

In parallel, there are smaller-scale local projects attempting to boost Guaraní NLP, where political considerations are limiting the growth of the project. Academics in the region have worked on Guaraní-Spanish rule-based translation and data augmentation techniques. For example, a 2024 study used grammar-based data augmentation to improve a neural Guaraní-Spanish translator, artificially generating Guaraní sentences from grammatical rules to supplement scarce training data (Lucas et al. 2024). This approach, led by researchers in Uruguay, achieved better translation accuracy by leveraging linguistic knowledge of Guaraní morphology (which is highly agglutinative). Such efforts underscore that Guaraní NLP is possible, but they often rely on a few experts and meagre funding. Unlike Irish or

Māori, there is less direct government investment in Guaraní language technology so far, despite its official status. Sociopolitical factors may offer an explanation here. While revered as a national symbol in Paraguay, Guaraní is associated more with oral tradition and rural life, and the tech sector remains Spanish-dominated. The risk in this context is thus that Guaraní might become an object of study or a checkbox for AI companies without empowering Guaraní communities to develop their own AI solutions. Moving forward, the Guaraní case would benefit from a model like Te Hiku's, e.g. a Paraguayan-led initiative, perhaps through universities or the Secretariat of Language Policy, to create language tools under local governance. There are promises, such as Guaraní translators emerging in open-source projects (like Apertium's rule-based engine, Forcada et al. 2011). The main takeaway is that without local control and investment, technology for Guaraní will remain piecemeal and at risk of exploitation. The community's voice needs to be central, or else Guaraní could be included in AI in form but not in spirit: the language will be reduced to a token presence, rather than a living, evolving language.

Last but not least, the Inuit languages of the Arctic have seen significant activity in language technology, offering a contrast of government-led and corporate initiatives. Inuktitut, spoken by roughly 40,000 Inuit in Canada (primarily in Nunavut and Nunavik), is noteworthy for having one of the largest Indigenous language corpora in the world. This is largely due to Canadian government practices. Since 1999, laws and legislative proceedings must be published in Inuktitut, leading to a substantial bilingual corpus of Inuktitut-English parliamentary records (the Nunavut Hansard). By 2018, researchers had compiled millions of parallel sentences from such sources (Kuhn et al. 2020). This availability of data enabled early MT projects. The National Research Council Canada (NRC) spearheaded an Indigenous Languages Technology project (2017–2020) that included an Inuktitut MT system, among other tools. The NRC project stands out for its empowerment-oriented approach. It was guided by an Indigenous advisory council and explicitly aimed to collaborate with communities, putting "equal emphasis on the agenda of the linguist and of the community" (Kuhn et al. 2020). Crucially, NRC made it a policy that it would not claim ownership of Indigenous language data collected or used in the project, acknowledging the "long, painful history of extractive research" where communities often lost access to their own linguistic data (Kuhn et al. 2020). By respecting the First Nations principles of OCAP, which assert that Indigenous peoples retain these rights over data about them, and partnering with Inuit organizations, the Canadian MT efforts sought to ensure that the tools (like an Inuktitut translator or a verb conjugator for Mohawk) met community needs and that the data remained accessible to those communities.

⁴ The Languages Law (Law 4251, enacted 2010) established the Secretariat for Language Policies and the Academy of the Guaraní Language, with the Academy specifically responsible for "standardizing the Guaraní language based on linguistic research" and "establishing the Guaraní alphabet" (Observatorio Legislativo CELE 2010).

In 2022–2024, Big Tech also entered the arena. Google Translate added Inuktitut (Inuktitut) to its supported languages, marking the first Canadian Indigenous language on the platform (News 2024). Google’s approach combined its massive neural architecture with consultation from Inuit representatives. According to Google, they worked with Inuit translators and the Inuit Tapiriit Kanatami (ITK, the national Inuit organization) to verify translation quality and address issues like the two different writing systems (Latin alphabet *qaliujaaqpait* and syllabics *qaniujaaqpait*). This consultation helped Google’s model handle Inuktitut text in both scripts. The inclusion of Inuktitut in a global service like Google Translate has clear benefits. Inuit people can now readily translate documents or websites, and non-speakers can access Inuktitut content, aiding everything from health services to cultural exchange. It also symbolically elevates Inuktitut’s status with an acknowledgement on the world stage of its importance. However, there are risks and questions that mirror those in the cases discussed earlier. One issue is quality and representation. Even with community input, Google’s system may generate incorrect or culturally inappropriate translations (especially given dialectal variation in Inuktitut). If users trust the AI too much, it could spread misinformation. Moreover, the matter of data ownership looms. Google’s model was undoubtedly trained on publicly available Inuit data, but now the model itself is proprietary to Google. Inuit communities do not have direct control over it. They cannot fine-tune Google’s translation model for local needs or prevent Google from using the data for other purposes. This is effectively a commercialization of a public resource: the years of translation done by the Inuit and the Canadian government have been harvested to create a product within Google’s ecosystem. This infringes on the ideal that Inuit should own and control their language data as per data sovereignty standards and practices. Indeed, as more Indigenous languages are added to big platforms, activists caution about a new wave of appropriation where respective languages might be used to enrich AI services without yielding commensurate investment back into those language communities. Additionally, there is a surveillance aspect: using Google Translate or other AI means interactions (text and voice) are logged on corporate servers, an evident data privacy concern. This raises trust issues, as some communities might prefer self-hosted tools to keep their data within their jurisdiction and under community governance.

Despite these concerns, the Inuktitut experience also offers a positive lesson in replicability. The existence of a large bilingual corpus was a game-changer. It shows the value of governments adopting policies (like bilingual records, or open data) that inadvertently create the raw material for language tech. Other Indigenous languages with official status (e.g. Māori and Irish) similarly benefit from translation mandates that produce parallel texts. These corpora

can seed LLM fine-tuning or MT systems. Furthermore, the collaborative model used by NRC to co-develop technology with Indigenous input at every stage is a best-practice template. It resulted in tools that communities actually wanted [such as a Kanyen’kéha (Mohawk) verb conjugator, which was specifically requested by an Indigenous educator (Canada NRC 2019)]. It also respected Indigenous rights by avoiding data extractivism. Under the OCAP framework, access and possession of data are as important as ownership, meaning communities must have the practical ability to get and use their data whenever they want. By not locking away the Inuktitut corpus or models, the Canadian projects empowered Inuit stakeholders to reuse and adapt them independently (Table 1).

4 Conclusion

The case studies above, alongside the Irish Gaelic experience, underline several common themes in applying LLMs and AI to Indigenous languages. First, community control and participation are crucial. Projects that centre the community, regardless of whether via governance boards, co-design workshops, or outright community ownership of data and models, tend to produce technologies that are more culturally attuned and sustainable. This observation is supported by Hogan-Brun & O’Rourke’s (2019) research about minority languages showing resilience, as the survival and growth of these languages stem from local initiative and valuing of diversity. When Māori data specialists or Inuit elders are part of AI development, the resulting system is built not only *with* the language but *of* the language, embedded in its cultural context. Second, there is a clear difference between relational and extractive ethics in data. A relational approach, inspired by Indigenous worldviews, treats language data as more than a commodity: it is a taonga or treasure, imbued with the soul of the people who speak it (Hernández 2025). We saw this with Te Hiku’s stance of “our voice, our data” and NRC’s refusal to claim Indigenous data. An extractive approach, by contrast, sees data as gold to be mined. This can lead to exploitation and a loss of trust. The CARE Principles for Indigenous Data have been proposed to complement the open data movement, and the above examples justify the necessity of the framework. Without ethical guardrails, even well-intentioned AI can end up recolonizing knowledge. Third, model replicability and accessibility emerged as both a hope and a challenge. The success of one language’s LLM (like UCCIX or Te Hiku’s ASR) can serve as a template, allowing adaptation by other languages with the right resources and expertise provided. Indeed, the open-source release of UCCIX explicitly aims to enable “adapting LLMs to other Indigenous languages” using its framework. Likewise, the methodologies in the

Table 1 Governance and ethical patterns in indigenous-language AI projects summary

Case	Lead	Aim (P/G/S)	Data governance stance	Big Tech role	Main risk	Mitigation
Irish (UCCIX and Ardntleacht)	UCCIX: University; Ardntleacht: State agency (Údarás na Gaeltachta)	P, S	Open releases; public oversight/advisory; public benefit licensing	Base model lineage (e.g., LLaMA-2); potential state vendors	Dependency or surveillance and cultural misfit	Open licensing, advisory governance, dialect targets
Māori (Te Hiku Media)	Community-owned media org	P, G	Māori data sovereignty; <i>kaitiakitanga</i> ; bespoke licenses; consent-first	Tooling under Māori terms (e.g., NVIDIA NeMo); no data surrender	Legal protection gaps for <i>taonga</i>	Community licenses, consent workflows, advocacy
Guaraní (Paraguay/Bolivia)	Platform addition (Google) and scattered local academic efforts	S	Limited platform governance; local projects use academic licenses	Central to Translate rollout; community role mainly evaluative	Tokenistic inclusion and orthography or dialect errors	Build local governance capacity; fund national corpora; community-run tools
Inuktitut (Canada)	Government research (NRC) and Indigenous advisory; later Google	P, S	OCAP-aligned policies; no ownership claims; corpora accessible	Significant in Translate; proprietary control	Enclosure or over-reliance and dialect/script bias	Co-development, keep corpora open, explore self-hosting

P, preservation; G, growth (revitalization); S, service

IBM research on Guaraní or the NRC's algorithms for poly-synthetic languages can be transferred. However, replicability faces practical hurdles. Many communities lack the funding, high-quality data, or expert teams to replicate the experience of Ireland or New Zealand. Solidarity and knowledge-sharing will therefore become important. Networks such as the Indigenous AI network and workshops (e.g., the ACL Workshop on Indigenous Languages) help spread these ideas. Finally, the cases highlight an evolving role for policy and governance. Governments can be enablers, but they need to work in true partnership with Indigenous communities to avoid top-down control. International bodies like UNESCO have also called for integrating Indigenous rights into AI development. This includes respecting rights declared in the UN Declaration on the Rights of Indigenous Peoples (UNDRIP), such as the right to use and transmit one's language and control one's cultural heritage. Implementing these rights in the AI space might mean requiring Free, Prior and Informed Consent (FPIC) before using a community's language data in a system, or ensuring that any AI product involving an Indigenous language has an oversight committee from that community.

In conclusion, the comparative perspective reinforces the central thesis emerging from the Irish case: LLMs for Indigenous and endangered languages hold great promise for revitalization, but realizing that promise without peril requires aligning AI development with the self-determination and values of the language communities. Whether it is Gaelic, Māori, Guaraní, or Inuktitut, the future of these and other Indigenous languages in the digital sphere should not be left to chance or solely to corporate algorithms. Instead, it calls for what Couldry and Mejias (2019) term a decolonial reversal, reclaiming the “datafied” future and ensuring that Indigenous peoples are not just data subjects but data authors and owners. By learning from each other's successes and missteps, and by centering principles like Indigenous data sovereignty, AI can realistically be harnessed for encouraging linguistic diversity and not against it.

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